

Binary Search :-

Start = 0, end = n-1;

$$\rightarrow \left(\text{mid} = \frac{\text{start} + \text{end}}{2} \right)$$

ceil value

$$\frac{28}{5}$$

$$\frac{a}{b} + 1$$

$$\left\lceil \frac{a+b-1}{b} \right\rceil$$

$$\frac{28+5-1}{5} = \frac{32}{5} = 6$$

Ques 1 - Find square root of a given number.

(return floor value)

$\text{int} \left(\frac{a}{b} \right) \rightarrow \text{Floor value}$

$\text{int} \left(\frac{28}{5} \right) \rightarrow 5 \dots \dots \rightarrow 5 \text{ only?}$

$$\frac{25}{5} = \frac{25+5-1}{5} = \frac{29}{5} = 5$$

$n=25$, $ans=5$
 $n=37$, $ans=6$

$$\sqrt{25}$$

1x1

2x2

3x3

4x4

5x5

6x6

7x7

```
int ans=1;  
for (int i=1; i<=n; i++) {
```

```
    if (i*i <= n)
```

```
        ans = i;
```

```
    else
```

```
        break;
```

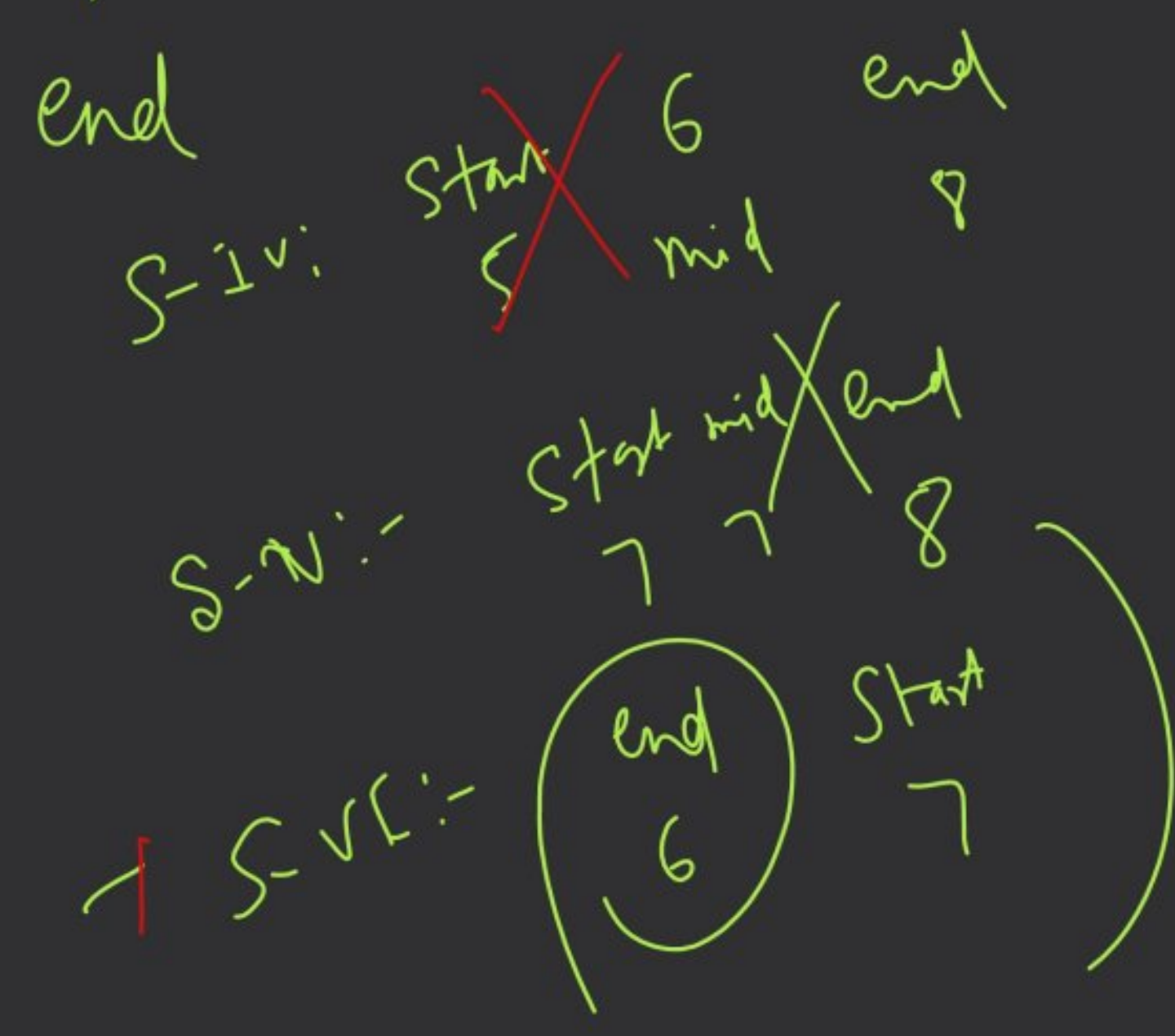
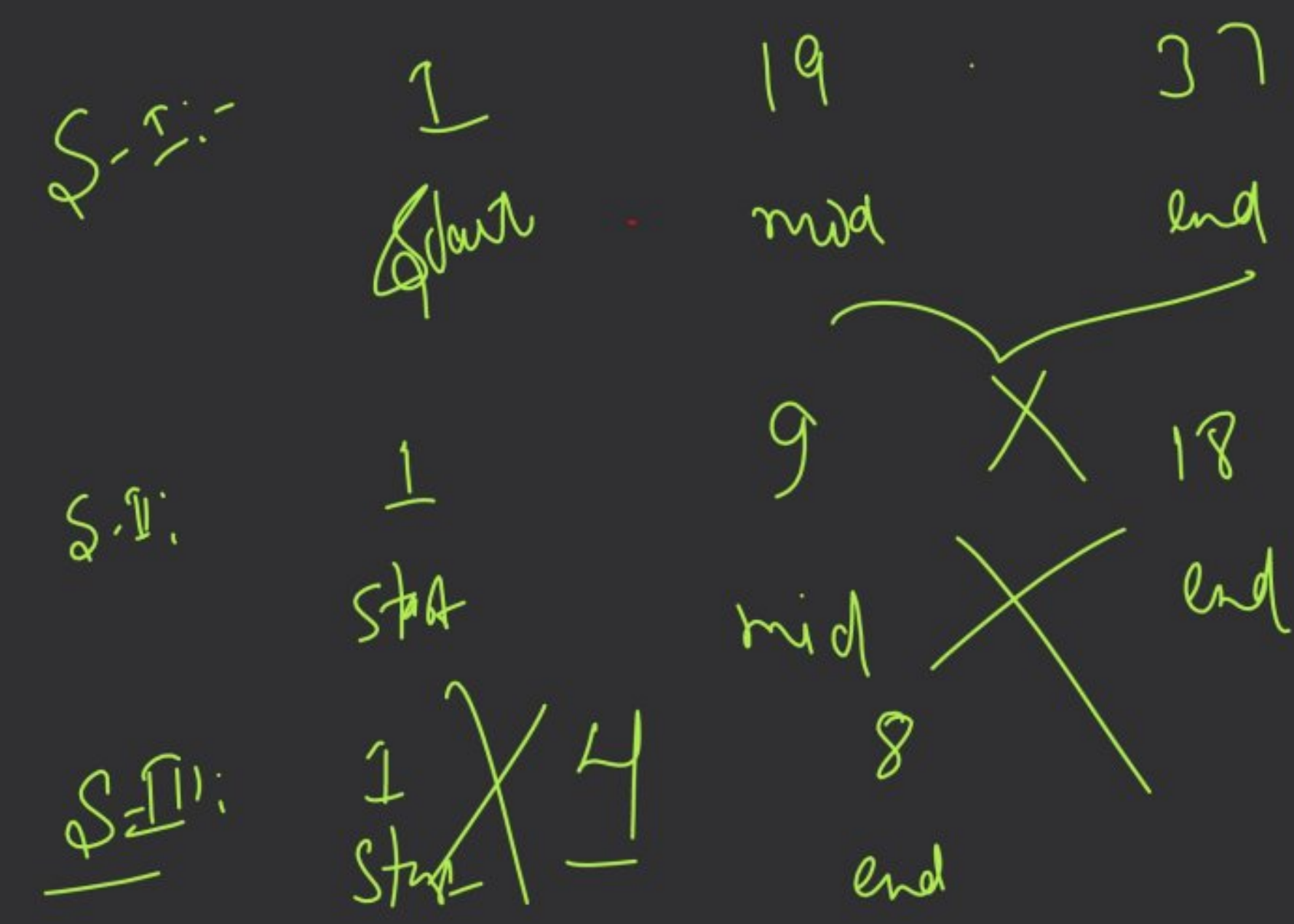
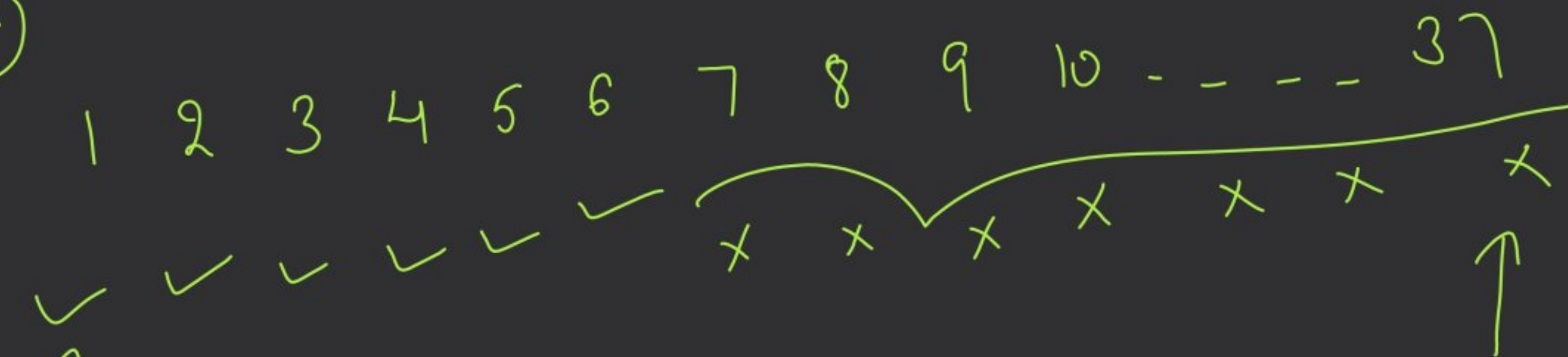
```
}
```

```
return ans;
```

$n=1$

③

ans ≥ 6



```
int start = 1, end = n, int ans = 1;
```

```
while (start <= end) {
```

```
return ans / end;
```

mid = 0 ←

```
int mid =  $\frac{\text{start} + \text{end}}{2}$ ;
```

```
if (mid * mid > n) {
```

```
end = mid - 1;
```

```
}
```

```
else {
```

```
start = mid + 1;
```

```
ans = mid;
```

```
}
```

```
return
```


piles = [3, 6, 7, 11] $\frac{7}{6}$

$\frac{3}{6} = 0$
 $\rightarrow 1 + 1 + 2 + 2$

possible

$h = 8$

not possible

→

	→ banana	X
1	2	3
X	X	X
3hr	2hr	1hr
6hr	3hr	2hr
7hr	4hr	3hr
11hr	5hr	4hr
11hr	5hr	4hr

it's impossible

✓	✓	✓
(4)	5	(6)
1hr	1hr	1hr
2hr	2hr	1hr
2hr	2hr	2hr
3hr	3hr	2hr
3hr	3hr	2hr
↓	↓	↓
8hr	8hr	11hr

✓	✓	✓	✓	✓
7	8	9	10	(11)
1hr	1hr	1hr	1hr	1hr
1hr	1hr	1hr	1hr	1hr
1hr	1hr	1hr	1hr	1hr
2hr	2hr	2hr	2hr	2hr
↓	↓	↓	↓	↓
5hr	5hr	5hr	5hr	5hr

time <= 8

$\checkmark$ $\times$ $\times$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$
 1 2 3 4 5 6 7 8 9 10 11
 $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 start end mid end end end end end end end

$\times$ $\times$
 1 3
 $\uparrow$ $\uparrow$
 start mid end
 4 5
 $\uparrow$ $\uparrow$
 start end
 not

ans = 4

3 4 7 1 1 1 1
 1 2
 $\frac{a}{b} = \frac{a+b-1}{b}$
 $\frac{pile + mid - 1}{mid}$

```
int start = 1; int ans = 1;
int end = maxPile(piles);
```

```
while (start <= end) {
    int mid =  $\frac{start + end}{2}$ ;
```

```
    long totalHr = 0;
```

```
    for (int pile : piles) {
```

```
        totalHr += (pile + mid - 1) / mid;
    }
```

```
    if (totalHr <= h) {
        ans = totalHr;
        end = mid - 1;
    }
```

```
    else
```

```
        start = mid + 1;
```

```
    }
```

```
    return start / ans;
```

$$2^{31} \quad 2^{31}$$

